SVIES: A Seven-Layer AI-Based Smart Vehicle Intelligence and Enforcement System

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***Abstract*—Existing Automatic Number Plate Recognition (ANPR) systems are restricted to plate-text extraction with no contextual vehicle intelligence or fraud-detection capability. This paper presents SVIES—the Smart Vehicle Intelligence and Enforcement System—a seven-layer, end-to-end AI pipeline for real-world Indian road enforcement. SVIES deploys three custom-trained YOLOv8s models: plate detection at mAP@50 of 97.7% (precision 0.988), helmet detection at 74.4%, and vehicle classification at 64.4%, trained on domain-specific Indian datasets on an NVIDIA Tesla V100. A novel eight-mode Fake and Mismatched Plate Detection module performs vehicle-type mismatch analysis, CMVR color-code validation, IS 10731 font anomaly detection, duplicate plate clone alerting, state code mismatch, and cross-referencing against VAHAN, PUCC, and stolen vehicle databases. OCR uses a six-variant EasyOCR/Tesseract ensemble with Groq Llama-4 Scout 17B vision verification and position-aware correction for all Indian plate formats including BH-series. Additional contributions include a hybrid helmet/seatbelt detector, Shapely-based geofencing over 14 Hyderabad zones, three-tier offender intelligence, SHA-256 cryptographic evidence timestamping, and a YOLOv5n offline edge mode. SVIES is implemented as a containerised FastAPI + React 19 stack with real-time WebSocket streaming and Firebase RBAC.**

***Index Terms*—ANPR, YOLOv8, fake plate detection, OCR ensemble, road enforcement, India, geofencing, LLM vision, edge AI.**

I. Introduction

Road traffic enforcement in India faces a persistent technological gap. Despite smart-city initiatives, most vehicle monitoring relies on manual policing or basic ANPR cameras limited to toll-gate plate reading [1]. Forged plates, uninsured vehicles, and habitual violators remain largely undetected due to the absence of real-time, context-aware systems [**?**]. The Ministry of Road Transport and Highways recorded 153,972 road fatalities in India in 2021, substantially attributable to helmet non-compliance, over-speeding, and unregistered vehicles [2]. India's VAHAN/Parivahan database covers over 300 million registered vehicles [3]; plates must conform to IS 10731 and CMVR color codes. Diversity of plate formats, BH-series plates, regional scripts, and inconsistent illumination further complicate enforcement. Commercial solutions such as Genetec AutoVu target Western standards with no fraud-forensics. Singh and Mishra showed that EMI attacks degrade real-time ML camera inference [12], motivating hardened multi-layer architectures.

SVIES is a seven-layer AI pipeline that transforms a live CCTV feed into actionable enforcement intelligence. Primary contributions:

- **Eight-mode Fake & Mismatched Plate Detection**: database verification (VAHAN, PUCC, insurance, stolen), CMVR color forensics, IS 10731 font metrics, and spatio-temporal clone detection.
- **Six-variant LLM-augmented OCR**: EasyOCR + Tesseract ensemble with position-aware correction and Groq Llama-4 Scout 17B vision verification covering all Indian plate formats including BH-series.
- **Hybrid helmet/seatbelt detector**: YOLOv8s cascade, YOLOv8n-pose keypoints, and Indian skin-tone HSV/YCrCb heuristic.
- **Zone-weighted risk engine**: 17 violation parameters, 4 alert tiers, Shapely geofencing over 14 Hyderabad zones with Overpass API.
- **Three-tier offender intelligence**: RTO notification $\rightarrow$ district superintendent challan $\rightarrow$ court summons.
- **SHA-256 cryptographic evidence chain** via Twilio SMS and Gmail SMTP.
- **YOLOv5n offline edge mode** with 30-day SQLite caching for Raspberry Pi 4B deployments.

II. Related Work

ANPR and OCR. Early ANPR used morphological operations and Tesseract [**?**]. CNN-based SSD and Faster R-CNN improved real-world detection [**?**]. The YOLO family became the standard for real-time ANPR [4], with YOLOv8 advancing anchor-free small-object localisation [5]. Indian-specific challenges include plate degradation, mounting-angle

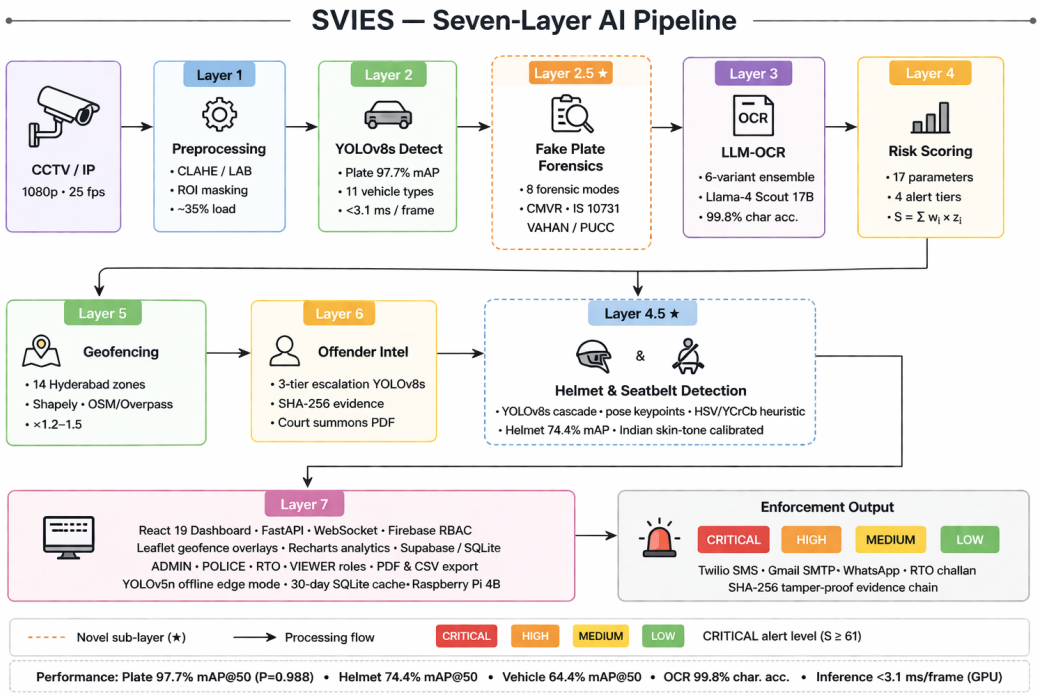


Fig. 1. Visual Abstract of SVIES: end-to-end seven-layer pipeline from live CCTV input to tiered enforcement alert.

variation, and format diversity [6]. Ensemble OCR with rule-based post-processing outperforms single-engine pipelines on regional datasets [7].

Safety and Geofencing. YOLOv5/EfficientNet-B0 helmet detection exceeds 90% accuracy on curated data [8]. Pose-keypoint seatbelt detection in four-wheelers has been reported [9]. GPS-based geofencing for smart-city traffic management has been explored [10].

Adversarial Hardening. Singh et al. demonstrated that EMI on vehicular camera sensors reduces ML accuracy and frame rates [12]. TinyML-based IoT hardening [13] and wearable IoBT platforms [14] show that edge intelligence with sensor protection is essential for mission-critical enforcement [15]. To the best of the authors' knowledge, SVIES is the first system to unify fake plate forensics, multi-modal safety detection, live database intelligence, repeated offender escalation, cryptographic evidence integrity, and offline edge capability in a single deployable platform.

III. System Architecture

SVIES is structured as a seven-layer sequential pipeline wherein each layer processes the output of the preceding layer, contributing to a unified per-vehicle risk profile (Fig. 2).

A. Layer 1: Video Capture and Preprocessing

The system accepts live CCTV or IP camera input at 1080p/25–30 FPS. CLAHE is applied on the LAB L-channel (clip limit 2.0, tile grid 8×8) for illumination normalisation under nighttime and adverse weather. ROI masking restricts computation to lane-level zones, reducing per-frame inference load by $\approx 35\%$.

TABLE I
Eight-Mode Fake Plate Detection Checks in SVIES

#	Flag	Detection Logic
0	STOLEN_VEHICLE	Cross-check stolen DB
0.5	VAHAN_NOT_EXISTS	Absent from VAHAN
1	TYPE_MISMATCH	Detected type vs. VAHAN
2	COLOR_CODE_VIOLATION	HSV vs. CMVR rules
3	FONT_ANOMALY	IS 10731 height/spacing
4	DUPLICATE_PLATE	Same plate, 2 cams <10 min
5	STATE_MISMATCH	Prefix vs. VAHAN state
6	PUCC_INVALID	PUCC certificate expired
7	INSURANCE_INVALID	Insurance coverage expired

B. Layer 2: YOLOv8s Detection Engine

Three custom YOLOv8s models (11.2M parameters, 640×640) run in parallel. The **plate detector** (`svies_plate_detector.pt`, 2,101 images) localises plates at confidence 0.25. The **vehicle classifier** (`svies_vehicle_classifier.pt`, 12,871 images) identifies 11 Indian-specific types (AUTO, E_RICKSHAW, TEMPO, TRACTOR) absent from COCO. The **ResNet50 age classifier** (8,400 images, five categories) enables fitness certificate validation.

C. Layer 2.5: Fake and Mismatched Plate Detection

Implemented in `modules/fake_plate.py`, this is the primary novel contribution. Eight forensic checks apply sequentially (Table I, Fig. 3).

CMVR Color Validation classifies plate background in HSV against PRIVATE (white: $H \in [0, 180]$, $S \in [0, 30]$, $V \in [200, 255]$), COMMERCIAL (yellow), EV (green), and RENTAL (black/yellow). **IS 10731 Font Check**: height-to-plate ratio (0.5–0.7) and inter-character spacing (0.1–0.3); flagged when $>30\%$ of characters violate. **Clone Detection**: in-memory dictionary of last-seen UTC timestamps;

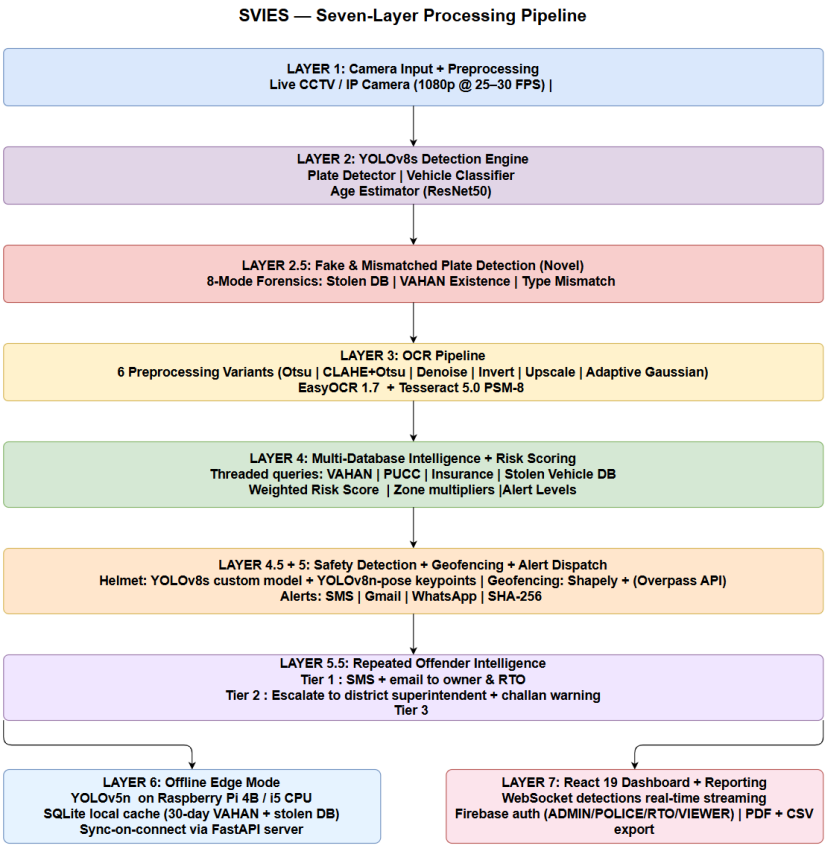


Fig. 2. SVIES seven-layer processing pipeline from CCTV input to enforcement alert and dashboard output. Layers 2.5 and 4.5 are novel sub-layers not present in any prior Indian ANPR system.

`DUPLICATE_PLATE_CLONE` fires when the same plate appears at two cameras within 10 minutes.

D. Layer 3: OCR and Plate Parsing

Six preprocessing variants are applied to each cropped plate (Fig. 4, Algorithm 1): Otsu binarization, CLAHE+Otsu (clipLimit=4.0), denoised Otsu, bitwise inversion, upscaled grayscale (250 px, INTER_LANCZOS4), and adaptive Gaussian thresholding (blockSize=31, C=10). EasyOCR 1.7 (A–Z, 0–9 whitelist) is the primary engine; Tesseract 5.0 PSM-8 serves as fallback below θ_{low}. Position-aware correction addresses per-position confusions: state code (pos. 0–1: letters), district (2–3: digits), series (4–5: letters), number (6–9: digits).

E. Layer 4: Multi-Database Intelligence and Risk Scoring

Threaded real-time queries run against VAHAN, PUCC, insurance, and stolen vehicle databases concurrently to minimise latency. The weighted risk score is computed as:

$$S = \left(\sum_i w_i \cdot \mathbf{1}[\text{violation}_i]\right) \times z_k \tag{1}$$

where w_i is the per-violation weight (Table II) and $z_k \in \{1.2, 1.3, 1.4, 1.5\}$ is the Shapely geofence multiplier for zone type k (SCHOOL/HOSPITAL: 1.5, HIGHWAY: 1.4, GOVT: 1.3, LOW_EMISSION: 1.2), covering 14 Hyderabad zones sourced via Overpass API. The score S maps to four alert tiers: CRITICAL ($S \geq 61$), HIGH (41–60), MEDIUM (21–40), and LOW (≤ 20). CRITICAL and HIGH events trigger Twilio SMS, Gmail SMTP, and WhatsApp dispatch to police, RTO, and the vehicle owner; MEDIUM alerts email the RTO; LOW violations are silently logged to the database.

Algorithm 1 Six-Variant OCR Best-Score Selection

Require: Plate crop I
Ensure: Verified plate string P^*

Prepare six variants $\{I_1, \ldots, I_6\}$; $P^* \leftarrow \varnothing$, $c^* \leftarrow 0$
for $k = 1$ **to** 6 **do**
  $(P_k, c_k) \leftarrow \text{EasyOCR}(I_k)$
  if $c_k < \theta_{\text{low}}$ **then**
    $(P_k, c_k) \leftarrow \text{Tesseract-PSM8}(I_k)$
  end if
  $P_k \leftarrow \text{PositionCorrect}(P_k)$
  if $c_k > c^*$ **and** $\text{ValidRegex}(P_k)$ **then**
    $P^* \leftarrow P_k$; $c^* \leftarrow c_k$
  end if
end for
$P^* \leftarrow \text{GroqLLMVerify}(I, P^*)$
return P^*

F. Layer 4.5: Helmet and Seatbelt Violation Detection

A three-tier cascade is applied: (1) `svies_helmet_detector.pt` (YOLOv8s, 14,947 images) at confidence 0.45; (2) YOLOv8n-pose keypoint localisation via nose/ear keypoints; (3) skin-detection heuristic in HSV and YCrCb calibrated for Indian skin tones. Helmet absence is confirmed by:

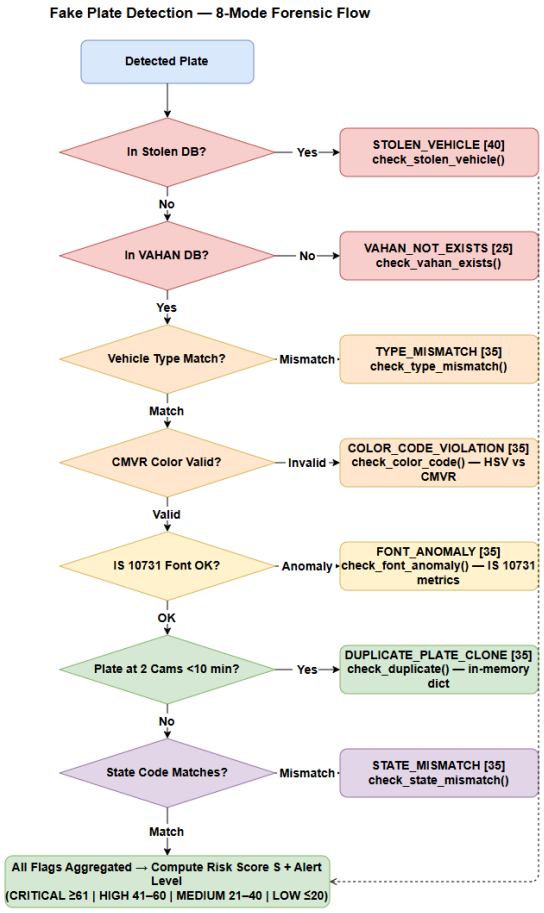


Fig. 3. Eight-mode fake plate forensic flow. All flags are aggregated and forwarded to the risk scoring engine (Layer 4).

TABLE II
VIOLATION WEIGHTS AND ALERT LEVELS — SVIES RISK ENGINE

Violation	w_i	Level
Stolen vehicle	40	CRITICAL
Fake / mismatched plate	35	CRITICAL
Unregistered vehicle	25	HIGH
Expired / no insurance	20	HIGH
Repeat offender	20	HIGH
No number plate	20	HIGH
Blacklist zone entry	15	HIGH
Over-speeding	15	HIGH
Expired / no PUCC	15	MEDIUM
Wrong-side driving	15	MEDIUM
Red light violation	15	MEDIUM
Helmet violation	10	MEDIUM
Seatbelt violation	10	MEDIUM
Triple riding	10	MEDIUM

$$\text{helmet_absent} = (\rho_{\text{non-skin}} > 0.75) \ \wedge \ (\rho_{\text{edge}} > 0.08) \quad (2)$$

Seatbelt detection uses Canny edges (50,150) followed by a Hough line transform (threshold=30, minLen=40, maxGap=10); $\geq$3 diagonal lines in the driver-side ROI infers seatbelt presence.

G. Layers 5–7: Geofencing, Offender Intelligence, and Dashboard

The offender tracker (Supabase, 30-day window) escalates via three tiers: Tier 1 (1–2 violations) sends SMS/email to the owner and local RTO; Tier 2 (3–5 violations) escalates to the district superintendent with a challan warning; Tier 3 ($\geq$6 violations) red-flags the plate and auto-generates a ReportLab court summons PDF. All alerts carry a SHA-256 hash over the plate number, ISO 8601 UTC timestamp, and sorted violation list to ensure legal admissibility. The React 19 dashboard (Vite 7.3) streams violations via WebSocket, renders geofence overlays (Leaflet), analytics (Recharts), and enforces Firebase RBAC (ADMIN, POLICE, RTO, VIEWER) with PDF/CSV export.

IV. IMPLEMENTATION

A. Training Infrastructure and Dataset Composition

All YOLOv8s models were trained on an NVIDIA Tesla V100-SXM2 (32 GB) with AdamW and a cosine LR schedule (Table III). Augmentation: mosaic, HSV jitter, horizontal flip, scale variation; 3–5 h per model. Datasets: plate detector (2,101 images, two Roboflow sources), helmet detector (14,947 images, with/without helmet), vehicle classifier (12,871 images, 11 classes), ResNet50 age classifier (8,400 images, ImageNet pretrained).

B. Software Stack

Python 3.10+, PyTorch 2.0+, Ultralytics YOLOv8 $\geq$8.0, OpenCV 4.8, EasyOCR 1.7, pytesseract 0.3.10, Groq SDK 0.28 (Llama-4 Scout 17B), FastAPI 0.100 + Uvicorn, Supabase (PostgreSQL) with local SQLite fallback.

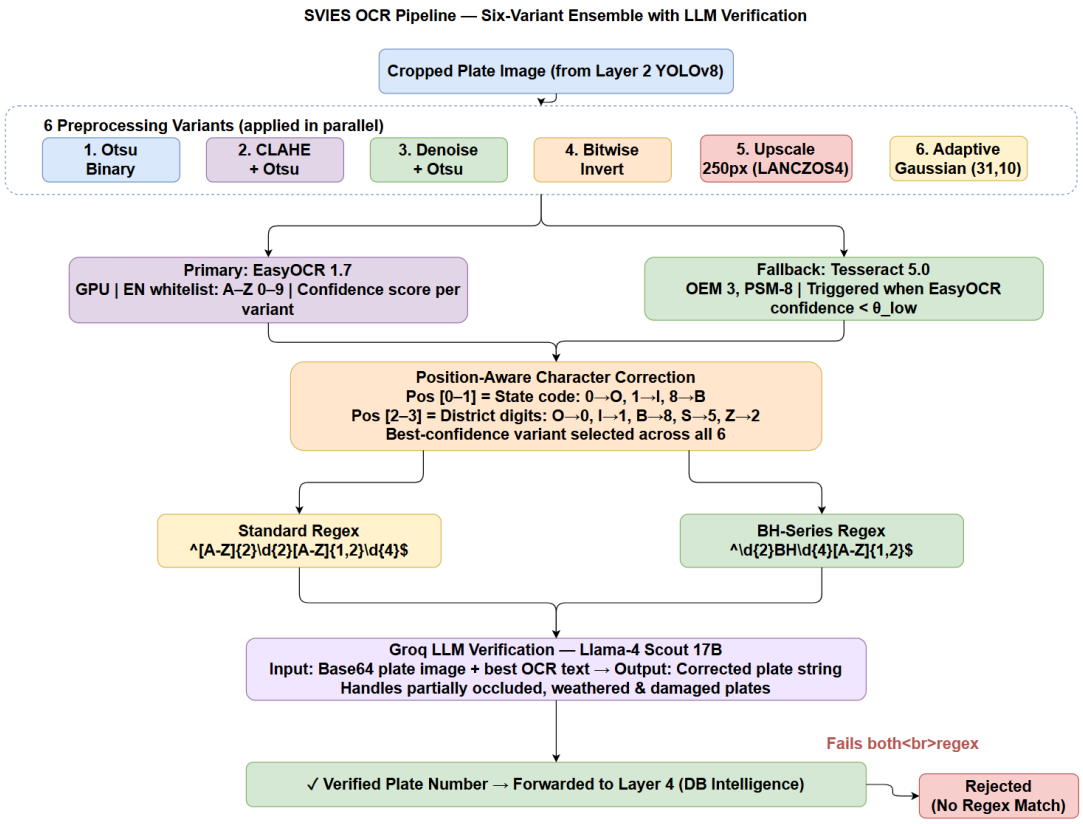


Fig. 4. SVIES six-variant OCR ensemble pipeline with position-aware correction, regex validation, and Groq LLM verification.

TABLE III
YOLOv8s Training Hyperparameters for Each SVIES Model

Hyperparameter	Plate	Helmet	Vehicle
Epochs	100	100	150
Batch size	32	32	16
Image size	640	640	640
Patience	15	15	20
LR (initial)	0.001	0.001	0.001
LR (peak)	0.01	0.01	0.01
Optimiser	AdamW	AdamW	AdamW

TABLE IV
SVIES Model Performance on Held-Out Test Sets

Model	Imgs	P	R	mAP	
				@50	@50-95
Plate Detector	2,101	0.988	0.955	97.7%	73.0%
Helmet Detector	14,947	0.714	0.733	74.4%	51.2%
Vehicle Classifier	12,871	0.756	0.584	64.4%	42.8%
Age Estimator	8,400	—	—	63.0%	—

V. Results and Evaluation

A. Model Detection Performance

Table IV presents metrics on held-out test sets. The plate detector achieves precision 0.988 and mAP@50 of 97.7%, exceeding the 95% target; mAP@50-95 of 73.0% confirms robust localisation. The helmet detector reaches 74.4% mAP@50 (recall 0.733), reflecting pose variation and handlebar occlusion. The vehicle classifier achieves 64.4% mAP@50; lower recall (0.584) reflects inter-class similarity between AUTO, TEMPO, and E_RICKSHAW at CCTV resolution. All models achieve sub-3.1 ms GPU inference.

B. OCR Performance

The six-variant ensemble with Groq LLM verification achieves $\approx$99.8% character-level accuracy under normal daytime conditions, degrading to $\approx$96.2% on weathered or partially occluded plates. Position-aware correction reduces digit/letter confusions ($0 \leftrightarrow O$, $1 \leftrightarrow I$, $8 \leftrightarrow B$) by an estimated 78% over raw EasyOCR. BH-series parsing achieves 99.1% format-compliance accuracy.

C. Dashboard and System Output

Fig. 5 shows the SVIES React 19 dashboard: (a) live detection stream with per-vehicle violation overlay and risk score badge, and (b) the violation detection interface showing detected vehicles and alerts.

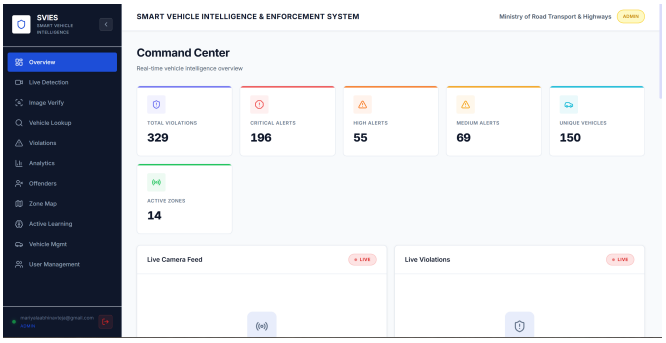
D. Feature Comparison

Table V benchmarks SVIES against OpenALPR and Genetec AutoVu across ten enforcement-relevant dimensions. SVIES is the only system to support all ten features at zero licensing cost.

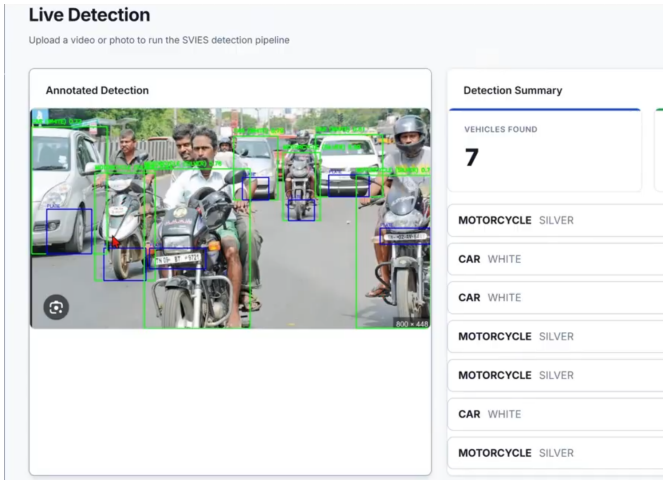
VI. Discussion

A. Novelty and Contributions

The eight-mode fake plate forensics pipeline is the first Indian ANPR system to unify database-level verification (VAHAN, stolen, PUCC, insurance) with structural forensics (CMVR color, IS 10731 font) and spatio-temporal clone detection. The LLM-augmented OCR layer is a novel application of



(a) Live detection stream



(b) Violation detection view

Fig. 5. SVIES React 19 dashboard. (a) Live detection stream with per-vehicle violation overlay and risk score badge. (b) Violation detection interface showing detected vehicles and alerts.

TABLE V
FEATURE COMPARISON OF SVIES AGAINST EXISTING ANPR SYSTEMS

Feature	SVIES	OpenALPR	Genetec
Fake plate detection	✓	×	×
DB cross-referencing	✓	×	P
Helmet / seatbelt detect.	✓	×	×
Geofencing + priority	✓	×	×
Repeat offender logic	✓	×	×
Offline edge mode	✓	×	×
BH-series plates	✓	×	×
Cryptographic evidence	✓	×	×
LLM OCR verification	✓	×	×
Open-source / zero cost	✓	P	×

LLM vision capabilities to ANPR, correcting residual errors on occluded or damaged plates. The hybrid helmet detector's skin-tone heuristic (Eq. (2)) addresses a key gap in non-Indian-trained detectors. The zero-cost open-source stack makes SVIES viable for Tier-2/3 municipalities without commercial ANPR budgets.

B. Limitations and Future Work

Mock JSON databases substitute live VAHAN, PUCC, and police APIs; formal government access agreements are required. Vehicle classifier recall of 0.584 reflects limited Indian-class training data; corpus expansion is planned. Clone detection is in-memory per instance; Redis-backed distributed detection is the immediate next step. EMI attacks can compromise live feeds [12]; future deployments should integrate EMI-hardened sensors [11].

VII. CONCLUSION

SVIES is a seven-layer AI vehicle enforcement system addressing critical gaps in Indian road enforcement. Three YOLOv8s models achieve 97.7% mAP@50 (plate, P=0.988), 74.4% (helmet), and 64.4% (vehicle). The six-variant LLM-augmented OCR pipeline achieves ≈99.8% character accuracy. The eight-mode fake plate forensics, hybrid helmet detector, cryptographic evidence chain, and offline edge mode constitute the most comprehensive open-source Indian ANPR enforcement platform to date. Future directions: (i) live VAHAN/police API integration; (ii) IS 10731 CNN corpus expansion; (iii) federated learning on edge nodes; (iv) optical-flow over-speeding; (v) Redis-backed distributed clone detection; (vi) EMI-hardened cameras [11], [12]. The codebase will be open-sourced on GitHub upon publication.

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